INSTITUTO SUPERIOR DE AGRONOMIA
Universidade de Lisboa

Impact of Environmental Stress on Crop Yield Across Sub-Saharan Africa

Analysis and Visualisation of Complex Agro-Environmental Data Final Project

Master's in Green Data Science

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1. Introduction

Drought and soil nitrogen deficiency are widely recognised as the two most important abiotic constraints to cereal and legume production in Sub-Saharan Africa (SSA), where the great majority of farming systems remain rainfed and low-input. Climate variability is increasing the frequency and severity of drought episodes across the region, while limited access to mineral fertiliser keeps soil nitrogen availability chronically low in most smallholder systems. Understanding how different crops and genotypes respond to these two stresses, individually, geographically, and over time, is essential for guiding variety recommendation, breeding priorities, and agronomic policy in the region.

The objective of this study is to evaluate the impact of major abiotic stresses (drought and nitrogen stress) on the yield performance of staple crops in SSA, and to determine which environmental and soil factors most strongly drive crop yield variation in the region.

1.1 Research Questions and Hypotheses

This project addresses two complementary research questions:

Question 1: Do drought and nitrogen stress significantly reduce grain yield in staple crops, and does the magnitude of this reduction vary by country and crop genotype?

Question 2: Does grain yield across crops respond systematically to environmental and soil characteristics, and if so, which factors matter most?

These translate into two working hypotheses:

H1 (stress response): Drought and Low-N stress significantly reduce grain yield in staple crops, with the magnitude of yield loss varying by country and crop genotype.

H2 (environmental drivers): Yield across staple crops responds to rainfall, temperature, and soil characteristics, with the strength of this relationship differing by crop.

2. Database Description

The dataset used in this project is the final project database created during the *Data Management Systems* course in the previous semester, comprising multi-environment crop breeding trials conducted in five Sub-Saharan African countries (Ethiopia, Ghana, Kenya, Nigeria and Tanzania) between 2020 and 2022. Trials were carried out by four breeding institutions: the Institute for Agricultural Research (IAR), the University of Ilorin, the International Maize and Wheat Improvement Center (CIMMYT), and the International Institute of Tropical Agriculture (IITA).

The dataset had already undergone cleaning and harmonisation during the construction of the database, including: standardising column structures across countries and crops into a unified template; standardising crop names, variety identifiers and trait labels to remove inconsistencies caused by spelling, abbreviations or local naming conventions; handling missing values and correcting inconsistent entries, including the treatment of outliers and invalid measurements; converting measurement units to ensure uniformity across datasets; removing duplicate or conflicting entries identified during cross-checking of multi-location and multi-year records; normalising environmental variables (rainfall, soil type, stress condition) for consistency; and creating placeholder columns for traits not measured in every crop, so that all crop datasets aligned to the same overall column structure prior to merging. This last step is the source of much of the missing data observed below:

traits specific to one crop (e.g. panicle measurements for Sorghum and Rice, or ear measurements for Maize) appear as missing for crops where that trait does not apply, rather than reflecting incomplete data collection.

Overall structure. The merged dataset contains 90,762 observations across 57 columns, covering four crops viz: Maize, Sorghum, Rice and Cowpea, with 407 unique genotypes in total (Table 1).

Table 1. Scope of the trial dataset by crop.

Crop	Rows	Genotypes	Countries	Regions	Management conditions
Maize	72,522	237	5	17	Optimum, Drought, Low-N
Sorghum	11,880	60	4	11	Optimum, Drought, Low-N
Rice	3,960	60	3	11	Optimum only
Cowpea	2,400	50	2	8	Optimum only

Coverage is not fully balanced across crop and country: Maize is the only crop present in all five countries, while Cowpea is restricted to Ghana and Nigeria, Rice to Ghana, Kenya and Nigeria, and Sorghum to Ghana, Kenya, Nigeria and Tanzania (none of the four crops include Ethiopia except Maize). Within each crop, however, the design is fully balanced: every genotype was tested an equal number of times across all combinations of region, year and management condition (where applicable), with two replicates per combination.

Key variables. Six continuous environmental and soil variables are recorded for every observation: rainfall (mm), mean temperature (°C), soil pH, and soil nitrogen, phosphorus and potassium content. These have negligible missingness (1 row out of 90,762, or 0.001%). The outcome variable, grain yield (kg/ha), has no missing values.

Missing data. Beyond the six environmental/soil variables and grain yield, the remaining columns are agronomic trait measurements (e.g. days to anthesis, panicle length, ear height, plant height) whose missingness is structural rather than incidental: each trait is only measured for the crop(s) to which it is physiologically relevant. This pattern is consistent across all crop-specific traits in the dataset and was already accounted for during the prior harmonisation of the database; it does not affect the variables used in the present analysis, all of which are fully populated.

Handling of Outliers: In all analyses, boxplot inspection of grain yield revealed numerous observations above the upper whisker (1.5 × IQR), particularly under Optimum management. These observations reflect the natural right-skewness typical of yield data rather than data errors, and the values fall within agronomically plausible ranges for the respective crops. Therefore, no outlier removal or trimming was applied in the final analyses, as the flagged observations represent genuine high-yielding genotype–location–year combinations that are relevant for identifying yield potential in each crop.

3. Exploratory Data Analysis

3.1 Maize: Stress Response and Genotype Performance

3.1.1 Effect of management condition on yield

Maize yield differed sharply by management condition (Figure 1). Mean yield fell from 3,042.9 kg/ha under Optimum management to 837.4 kg/ha under Drought (a 72.5% reduction) and to only 394.3 kg/ha under Low-N (an 87.0% reduction). A one-way ANOVA confirmed this effect was highly significant and substantial in magnitude ($F = 17{,}656.6$, $p < 0.001$, $\eta^2 = 0.328$). Tukey's HSD test confirmed that all three conditions differ significantly from one another (all $p < 0.001$), and that Low-N caused significantly greater yield loss than Drought (mean difference = 443.1 kg/ha). This pattern held in every one of the five countries, though its magnitude varied considerably (Table 2). Kenya showed the most severe Drought penalty (81.4% loss) while Ghana showed the smallest (59.5%); Kenya also showed the most severe Low-N penalty (93.4%) while Tanzania showed the smallest (70.3%) (Figure 2 and Table 2).

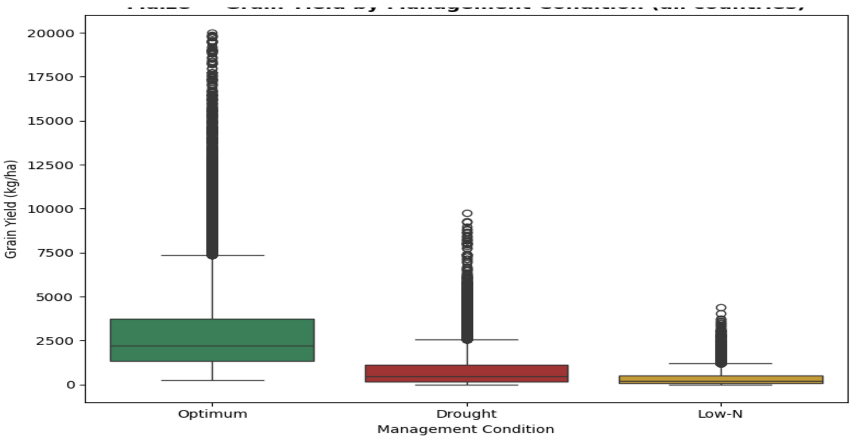

Figure 1. Maize grain yield distribution by management condition (all countries combined).

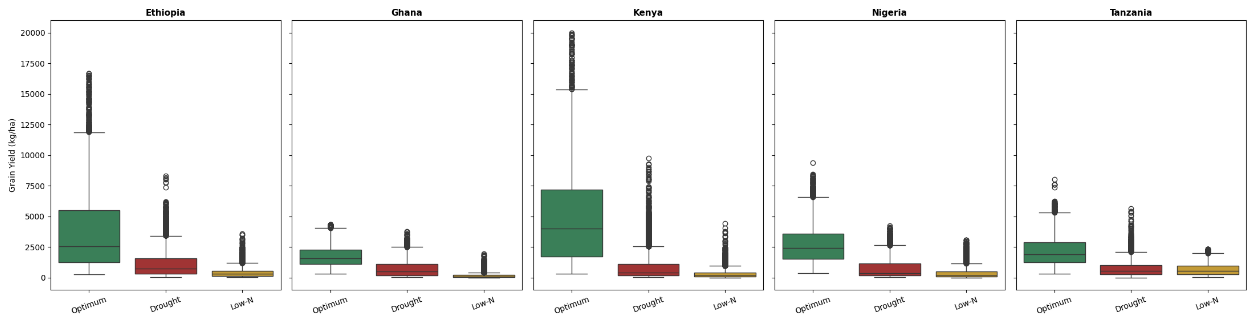

Figure 2. Maize grain yield by management condition, faceted by country.

Table 2. Maize: per-country ANOVA results for management condition effect on yield (all p < 0.001).

Country	F-statistic	η^2	% loss Drought	% loss Low-N
Ethiopia	3,165.7	0.331	69.8%	88.8%
Ghana	7,037.3	0.524	59.5%	90.4%
Kenya	4,604.7	0.419	81.4%	93.4%
Nigeria	10,462.8	0.495	72.7%	85.2%
Tanzania	4,414.6	0.408	66.7%	70.3%

3.1.2 Region × Management interaction

A two-way ANOVA of yield against Region, Management Condition, and their interaction confirmed that the size of the stress penalty depends on location. Management condition remained the dominant single factor ($\eta^2 = 0.328$), but the Region × Management interaction was also substantial ($\eta^2 = 0.211$), exceeding the main effect of Region alone ($\eta^2 = 0.158$). Optimum yields vary considerably across the seventeen regions (peaking above 8,500 kg/ha in Kitale and Bako), Drought and Low-N yields converge towards a much lower, narrower band across most regions, with a small number of regions (notably Kitale, Zaria and Bako) showing comparatively better-than-average resilience under stress (Figure 3).

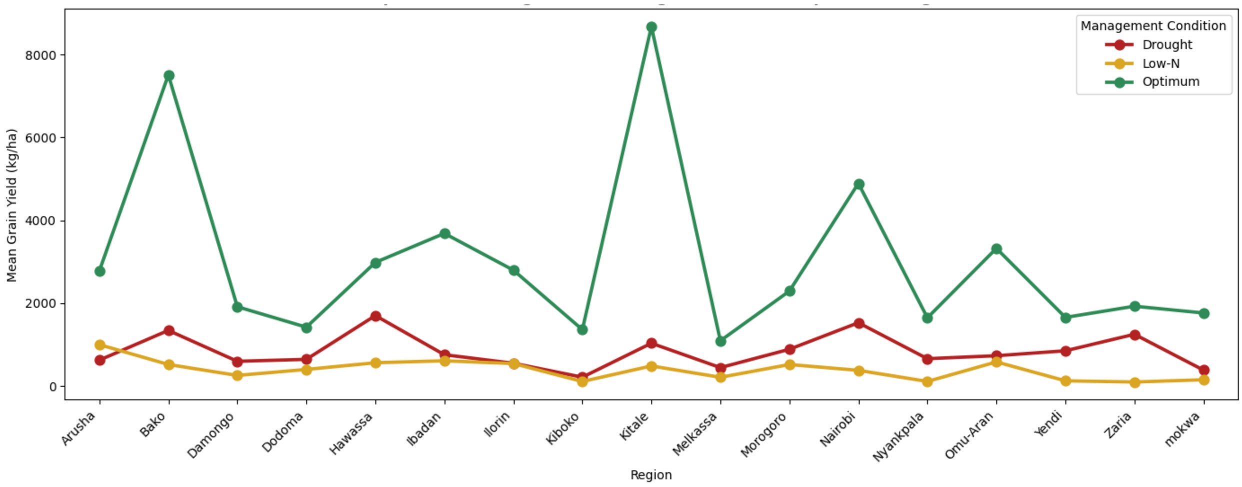

Figure 3. Region × Management Condition interaction plot — Maize.

3.1.3 Genotype clustering

Ward-linkage hierarchical clustering of the 237 Maize genotypes, based on their mean yield across all Region × Management combinations, identified three clusters at k = 3 (figure 4). Cluster 1 (n = 20) showed moderate Optimum yield but the best relative retention of yield under stress. Cluster 2 (n = 56) achieved the highest Optimum yield (4,128.8 kg/ha) but suffered the steepest proportional decline under both stresses, while Cluster 3 (n = 161, the majority) showed intermediate-to-low performance throughout (figure 5).

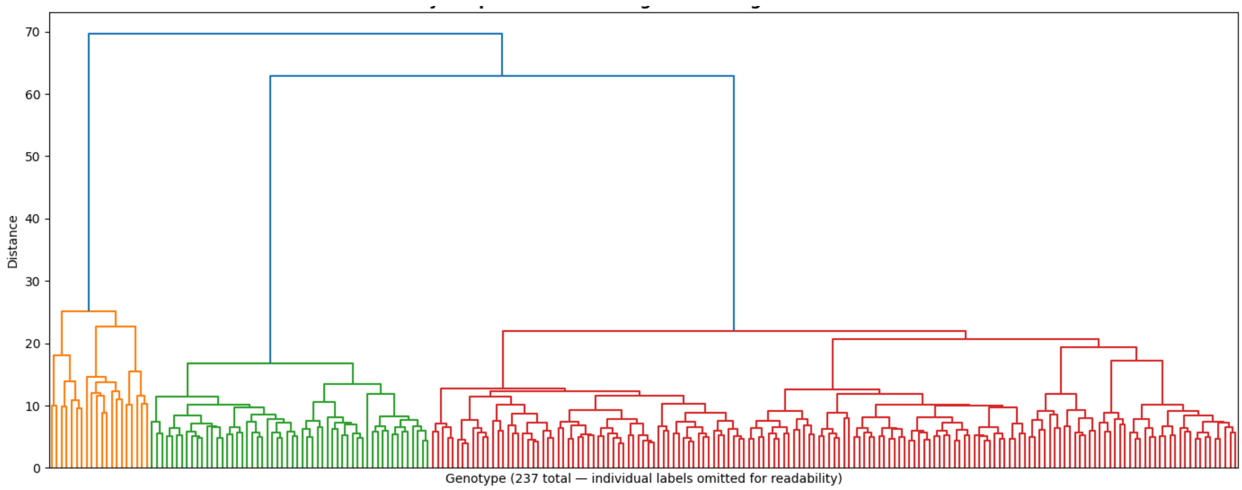

Figure 4. Dendrogram of Maize genotype clustering (Ward linkage).

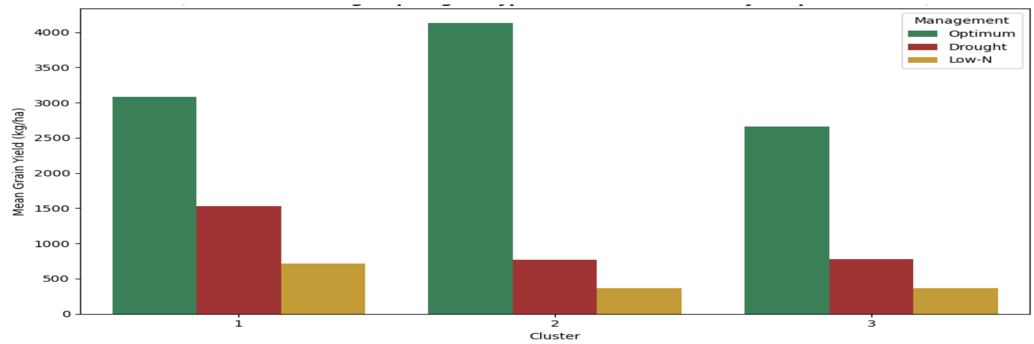

Figure 5. Maize genotype cluster mean yield profiles across the three management conditions.

3.2 Sorghum: Stress Response and Genotype Performance

3.2.1 Effect of management condition on yield

Sorghum showed a markedly different stress response profile to Maize. Mean yield fell only modestly from Optimum (5,047.6 kg/ha) to Drought (4,105.2 kg/ha), an 18.7% reduction, but collapsed sharply under Low-N (922.1 kg/ha, an 81.7% reduction) Table 3. A one-way ANOVA confirmed both effects were highly significant overall ($F = 4,526.3$, $p < 0.001$, $\eta^2 = 0.433$).

Table 3. Sorghum grain yield summary statistics by management condition (all countries combined).

Condition	n	Mean (kg/ha)	SD	Min	Max
Optimum	3,960	5,047.6	2,658.4	555.0	15,740.3
Drought	3,960	4,105.2	2,216.0	349.6	14,566.9
Low-N	3,960	922.1	536.5	90.6	3,973.4

3.2.2 Region × Management interaction

The per-country breakdown shows that drought losses ranged from a negligible 8.6% (Kenya) to a small yield gain under Drought relative to Optimum in Nigeria, where Drought yields were 6.2% higher than Optimum, while low-N losses remained severe and consistent across all four countries (73.8%–84.7%). The Region × Management interaction was comparatively small ($\eta^2 = 0.045$) relative to the Management main effect ($\eta^2 = 0.433$). Optimum yields vary considerably across the eleven regions (peaking above 4,500 kg/ha in Nairobi). Drought yields track optimum yields fairly closely across most regions, with the two lines often running near-parallel and even crossing in places such as Dodoma and Omu-Aran. Low-N yields, by contrast, remain compressed near the bottom of the range in every region, regardless of how high optimum or drought yields climb (Figure 6).

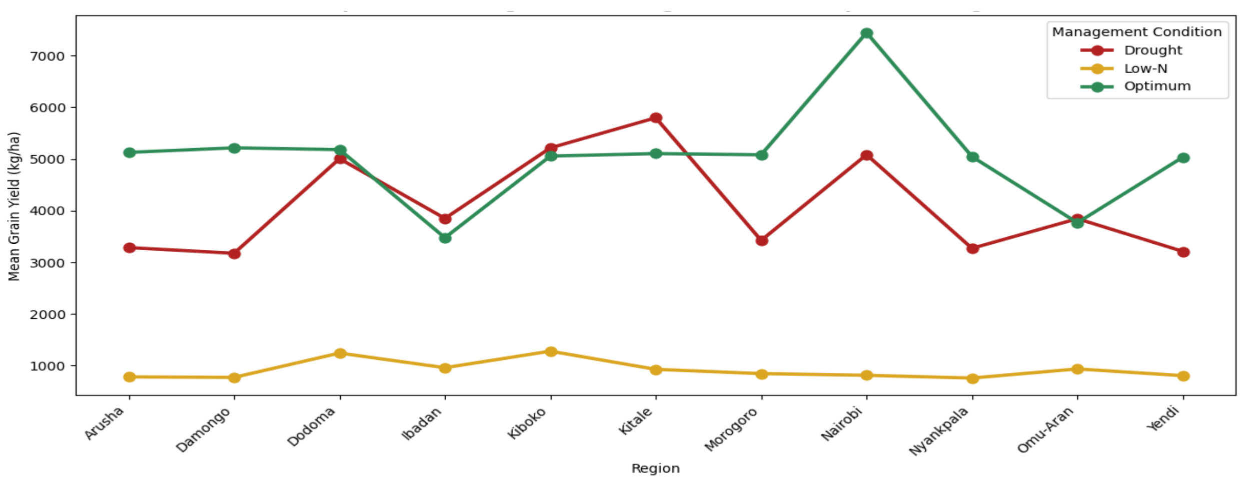

Figure 6: Region × Management Condition interaction plot — Sorghum.

3.2.3 Genotype clustering

Genotype clustering (Ward linkage, k = 3) on the 60 Sorghum genotypes identified three clusters (Figure 7). Cluster 1 (n = 22) was both the highest-yielding cluster under Optimum conditions (7,169.2 kg/ha) and the most resilient under Drought. Cluster 2 (n = 6) was consistently the lowest-yielding group across all conditions, while Cluster 3 (n = 32, the largest cluster) showed intermediate performance (Figure 8).

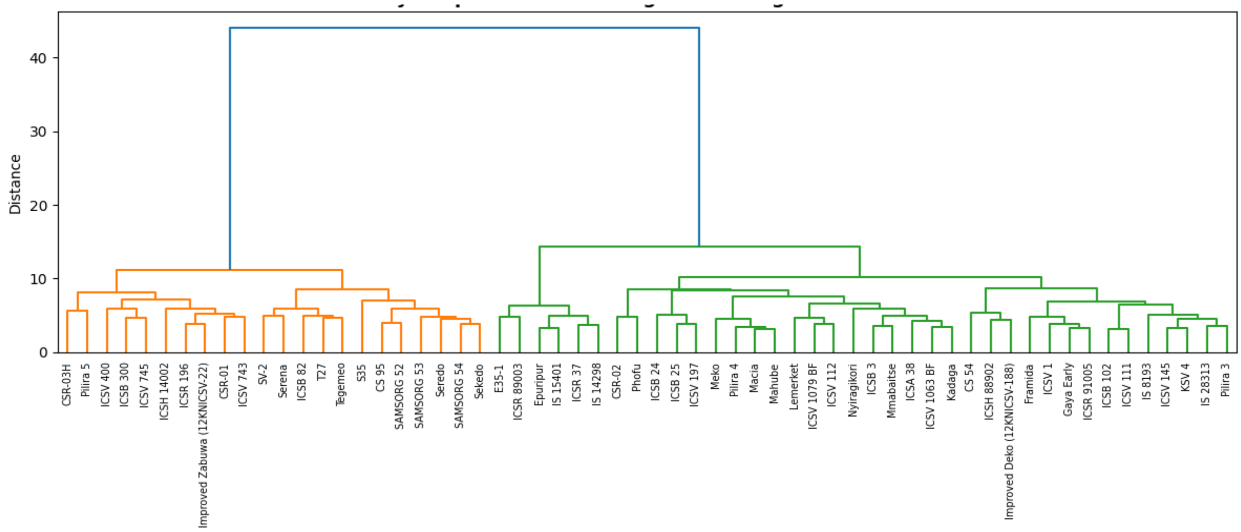

Figure 7: Dendrogram of Sorghum genotype clustering (Ward linkage).

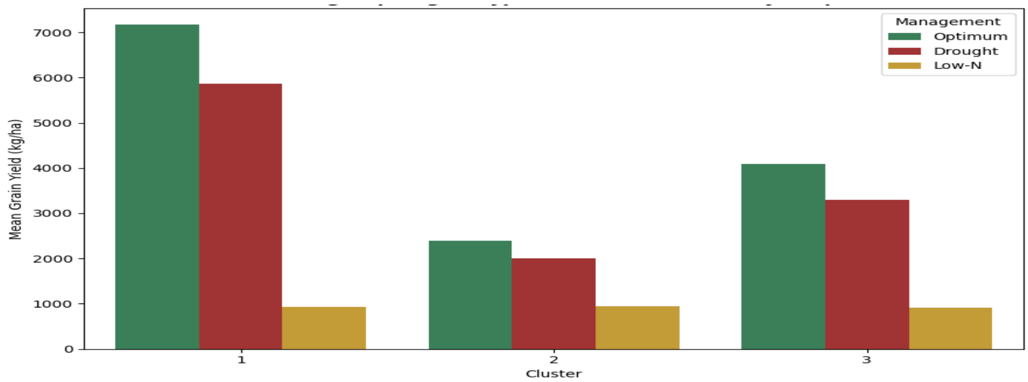

Figure 8. Sorghum genotype cluster mean yield profiles across the three management conditions.

3.3 Cowpea: Genotype Performance

Cowpea yield differed significantly by country ($F(1, 2398) = 53.4$, $p < 0.001$), though the effect size was small ($\eta^2 = 0.022$). Nigeria recorded higher mean yield (2,746.6 kg/ha) than Ghana (2,207.1 kg/ha) (Figure 9a). Year had negligible effect (means of 2,535.4, 2,541.7, and 2,555.8 kg/ha across 2020–2022) (Figure 9b).

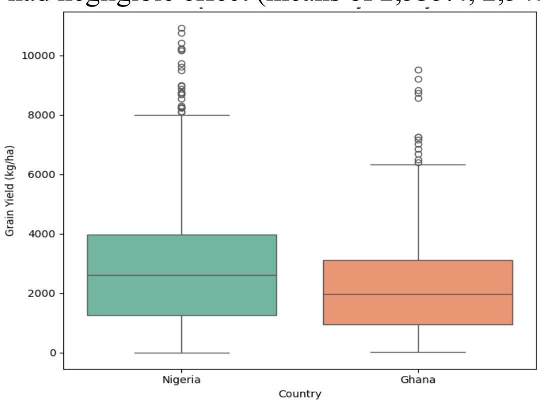

Figure 9a: Cowpea grain yield by country

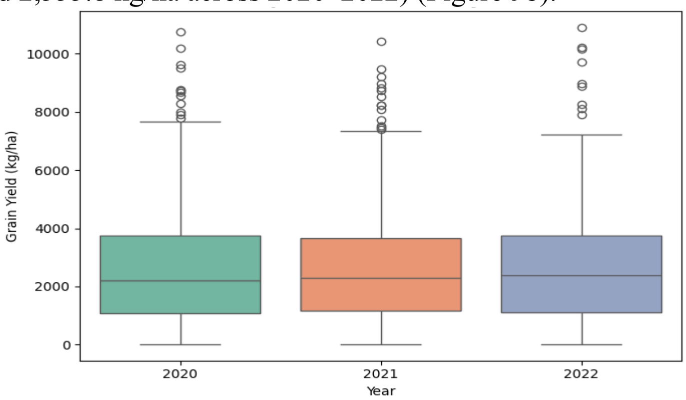

Figure 9b: Cowpea grain yield by Year

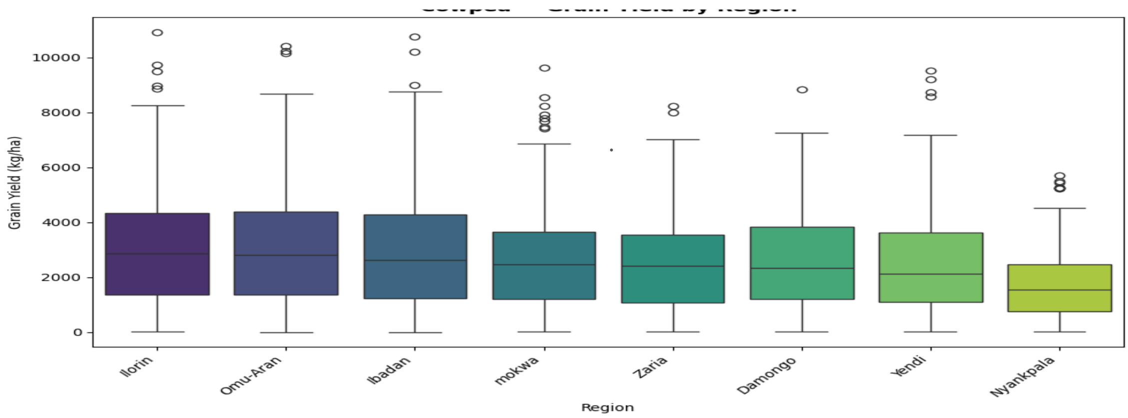

Figure 9c. Cowpea grain yield distribution by region

Genotype clustering (Ward linkage, k = 2) on the 50 Cowpea genotypes identified 2 clusters (Figure 10). Cluster 1 (n = 5) was the High-yielding tier (mean 4419.3 kg/ha) and the second cluster comprises the remaining genotypes, forming the Low-yielding tier (n = 45; mean 2,336.0 kg/ha).

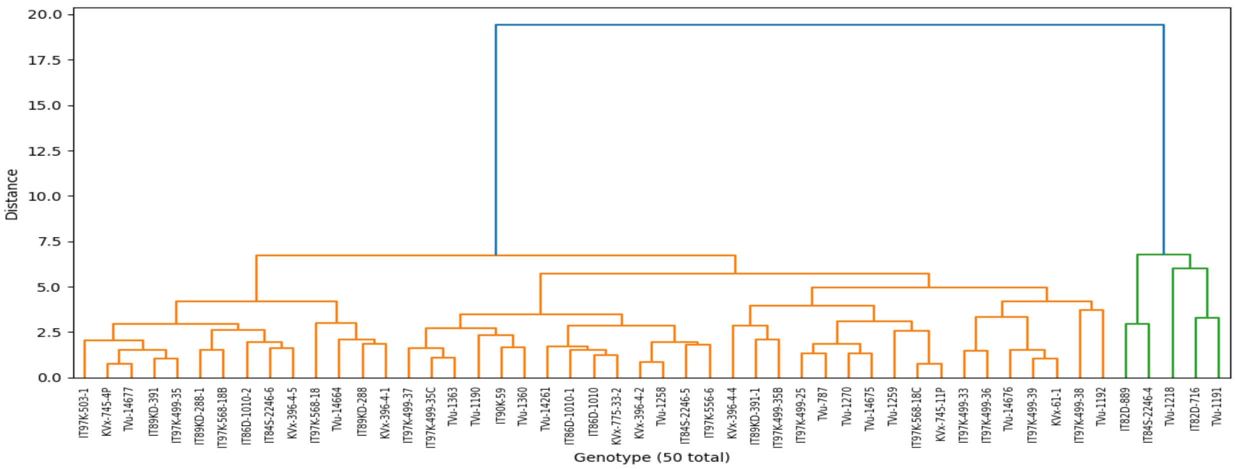

Figure 10. Dendrogram of Cowpea genotype clustering (Ward linkage), based on yield pattern across regions

3.4 Rice: Genotype Performance

Rice yield varied significantly by country ($F(2, 3957) = 192.3$, $p < 0.001$, $\eta^2 = 0.089$). Kenya recorded substantially higher mean yield (5,096.2 kg/ha) than Nigeria (3,386.8 kg/ha) or Ghana (3,176.8 kg/ha) (11a). As with the other crops, year effects were minor for Rice (means ranging 3,745.0–3,835.7 kg/ha) (Figure 11b).

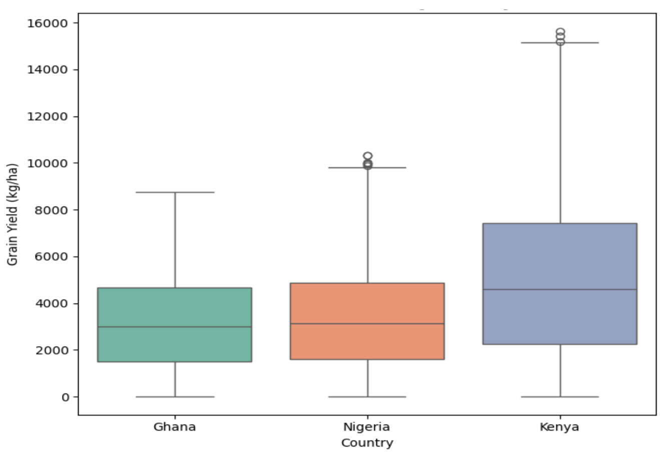

Figure 11a: Rice grain yield by country

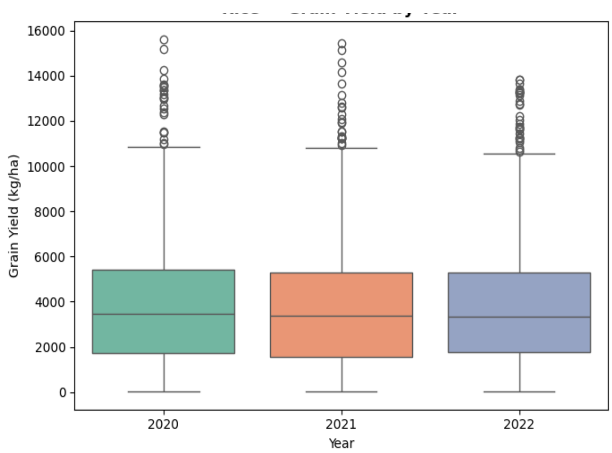

Figure11b: Rice grain yield by Year

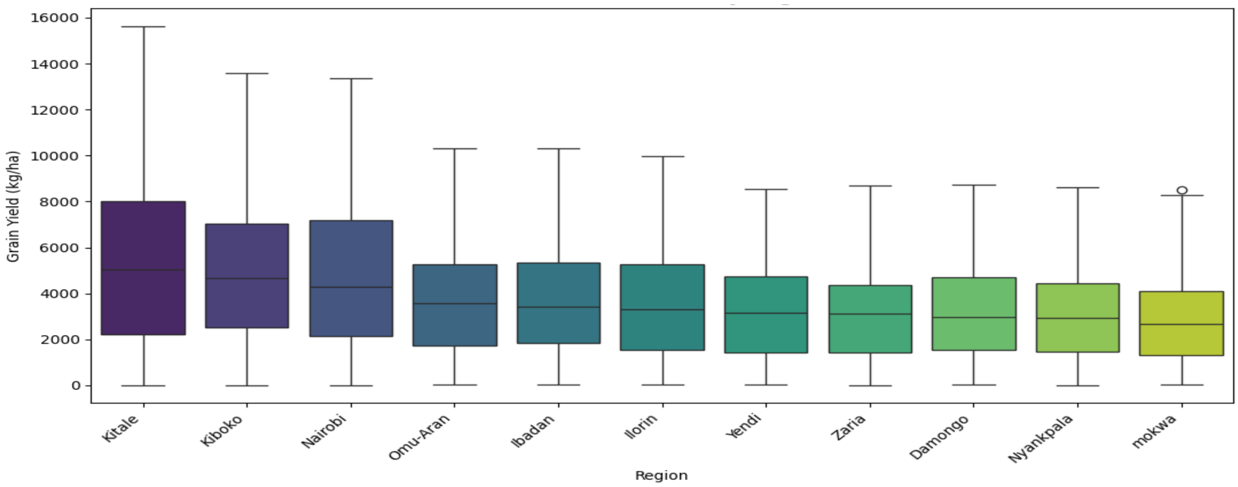

Figure 11c. Rice grain yield distribution by region

Genotype clustering for Rice produced two clusters, a High-yielding tier (n = 16, mean 5,421.1 kg/ha) and a Low-yielding tier (n=44, mean 3,204.7kg/ha) (figure 12)

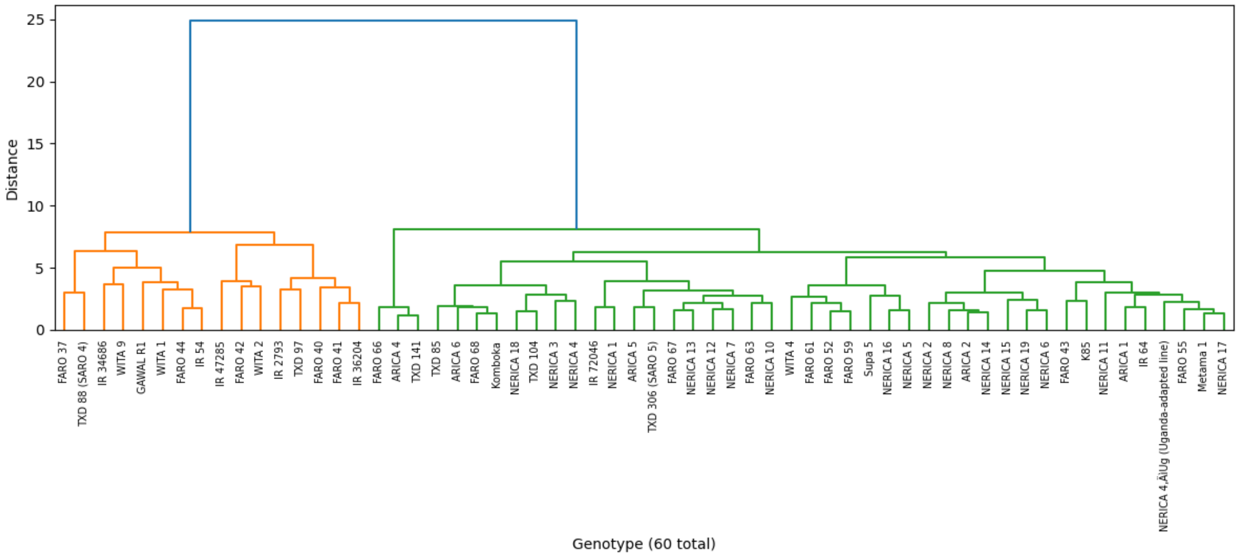

Figure 12. Dendrogram of Rice genotype clustering (Ward linkage), based on yield pattern across regions

3.5 GGE Biplot Analysis of Genotype-by-Environment Interaction

The GGE biplot of Maize genotypes across Country × Management environments explained 81.3% of total variation (PC1 = 71.7%, PC2 = 9.6%) (Figure 13a). Genotypes separated clearly along PC1, with two distinct groups: a cluster of high-yielding genotypes (mean yield 1,700–2,300 kg/ha) positioned on the right side of the biplot, and a larger cluster of lower-yielding genotypes (mean yield below 1,400 kg/ha) on the left. SAMMAZ-52, ILOMAZ-61, ILOMAZ-77, ILOMAZ-83, and ILOMAZ-84 were positioned furthest along PC1 and recorded the highest mean yields in the dataset. The Optimum-condition environment vectors (Kenya_Optimum, Ghana_Optimum, Nigeria_Optimum, Tanzania_Optimum, Ethiopia_Optimum) pointed toward this high-yielding genotype cluster. By contrast, the Drought and Low-N environment vectors for all five countries pointed toward the origin and into the upper-left quadrant, away from the high-yielding genotype cluster, with comparatively short vector lengths relative to the Optimum vectors.

The Sorghum GGE biplot explained 96.1% of total variation (PC1 = 94.4%, PC2 = 1.7%), with genotype separation almost entirely captured along PC1 (Figure 13b). Genotypes positioned furthest along the positive

PC1 axis (Sekedo, S35, SAMSORG 53, SAMSORG 54, SAMSORG 52, Tegemeo, and T27) recorded the highest mean yields (3,500–4,900 kg/ha), while genotypes on the negative PC1 axis recorded the lowest mean yields (1,500–2,500 kg/ha). The Drought environment vectors (Ghana_Drought, Kenya_Drought, Tanzania_Drought, Nigeria_Drought) extended furthest from the origin and pointed toward the negative PC2 region, in the direction of genotypes including Serena, SV-2, Seredo, and Improved Zabuwa. The Optimum environment vectors (Nigeria_Optimum, Tanzania_Optimum, Kenya_Optimum) pointed in a similar but less extreme direction. The Low-N environment vectors for all four countries clustered tightly near the origin with markedly shorter vector lengths than either the Drought or Optimum.

The Rice GGE biplot, constructed using Country × Year as the environment definition, explained 79.6% of total variation (PC1 = 67.6%, PC2 = 12.0%) (Figure 13c). Genotypes separated along PC1, with IR 2793, FARO 40, FARO 41, FARO 42, FARO 44, IR 34686, IR 54, and IR 36204 positioned on the positive PC1 axis and recording the highest mean yields (4,500–6,500 kg/ha). WITA 1 was notably separated along PC2, positioned furthest in the direction of the Kenya_2022 environment vector, which itself extended furthest from the origin among all environment vectors and pointed in a distinct direction (upper region of the biplot) relative to the remaining environment vectors. The Nigeria, Ghana, and Kenya_2021 environment vectors for 2020–2022 clustered closely together pointing toward the lower-right quadrant, in the direction of FARO 37, FARO 40, WITA 2, and TXD 88 (SARO 4), while GAWAL R1 was positioned in the opposite direction (negative PC2), away from all environment vectors.

The Cowpea GGE biplot, constructed using Country × Year as the environment definition, explained 78.9% of total variation (PC1 = 69.5%, PC2 = 9.4%) (Figure 13d). Genotypes separated along PC1, with TVu-1191, IT82D-716, TVu-1218, IT84S-2246-4, and IT82D-889 positioned on the positive PC1 axis and recording the highest mean yields, while the remaining genotypes clustered tightly together on the negative PC1 axis with lower mean yields. The Ghana_2021 and Ghana_2022 environment vectors pointed toward the upper-right quadrant, in the direction of TVu-1218, while the Nigeria_2020 and Nigeria_2021 environment vectors pointed toward the lower-right quadrant, in the direction of IT84S-2246-4 and IT82D-889. The Ghana_2020 environment vector was notably shorter than the other environment vectors.

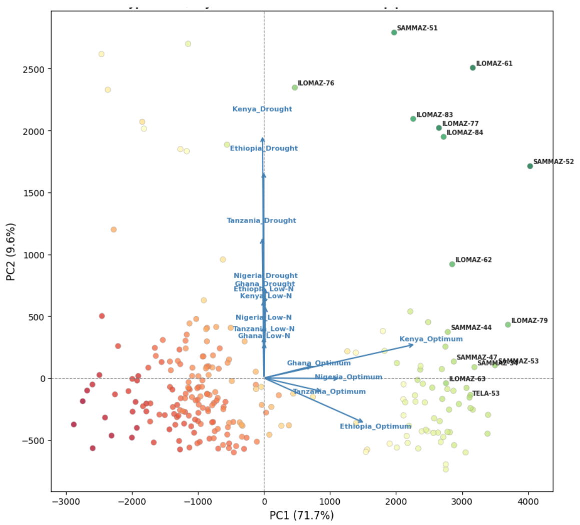

Figure 13a: GGE Biplot of maize genotypes

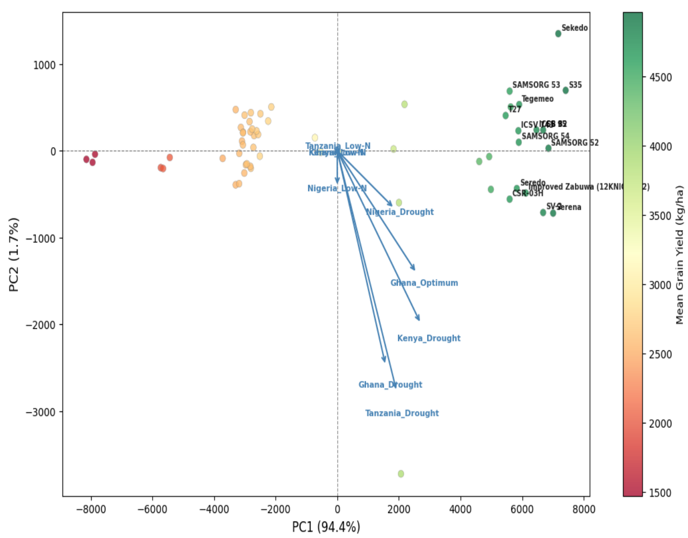

Figure 13b: GGE Biplot of Sorghum genotypes

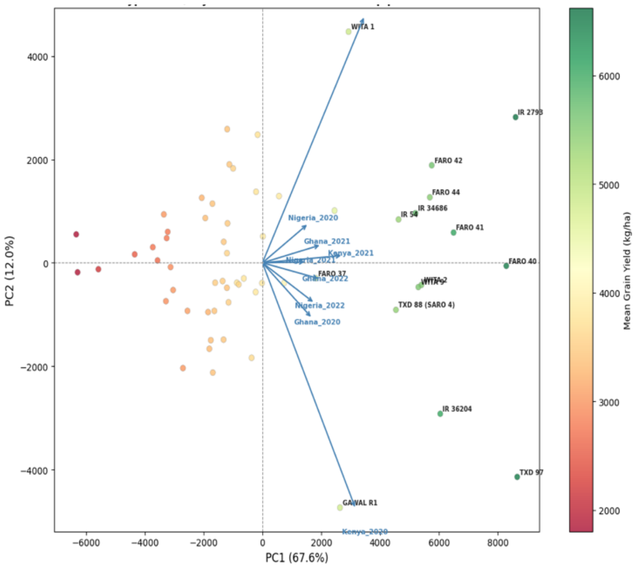

Figure 13c: GGE Biplot of Rice genotypes

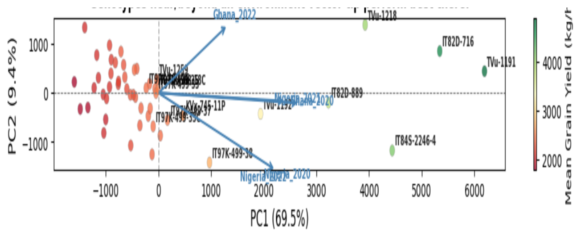

Figure 13d: GGE Biplot of Cowpea genotypes

3.6 Environmental and Soil Drivers of Yield

Multiple linear regression of grain yield on six environmental/soil predictors (rainfall, mean temperature, soil pH, soil N, soil P, soil K) was performed separately for each crop. Explanatory power varied considerably (Table 4).

Table 4. Multiple linear regression of grain yield on environmental and soil variables, by crop

Crop	R^2	Adjusted R^2	F-statistic	n
Maize	0.527	0.527	13,451.4	72,521
Sorghum	0.447	0.447	1,598.9	11,880
Rice	0.096	0.094	69.5	3,960
Cowpea	0.043	0.041	17.9	2,400

Environmental and soil variables explained the majority of yield variance in Maize ($R^2 = 0.527$) and a substantial share in Sorghum ($R^2 = 0.447$), but explained yield variance poorly in Rice ($R^2 = 0.096$) and Cowpea ($R^2 = 0.043$) (Figure 14).

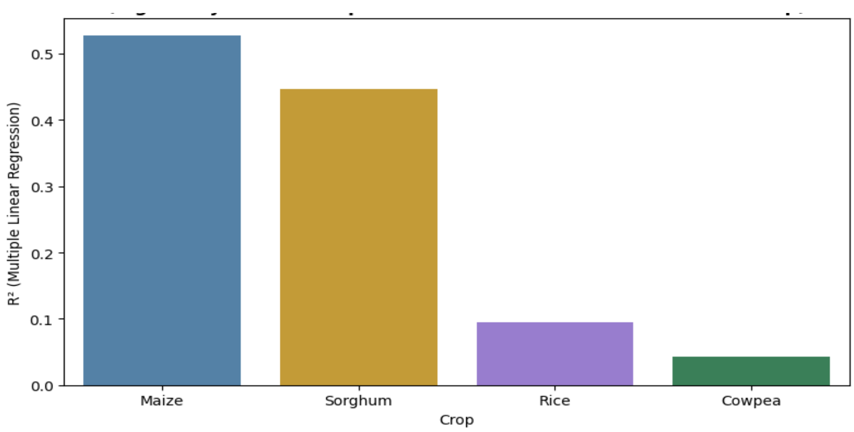

Figure 14. Proportion of yield variance explained by environment/soil variables (R^2), by crop.

Variance Inflation Factor (VIF) diagnostics revealed substantial multicollinearity among predictors in all four crops, most severely in Cowpea (soil pH VIF = 3,890; rainfall VIF = 3,319) and Rice (soil P VIF = 1,355), and to a lesser extent in Sorghum and Maize (maximum VIF = 651 and 108 respectively). Standardised regression coefficients (Table 5) were used to compare the relative influence of each variable within and across crops, despite the collinearity caveat above.

Soil nitrogen content was the strongest single predictor of yield in three of the four crops (Maize, Sorghum, Rice), and the second-strongest in Cowpea. Rainfall showed an unexpected negative association with yield in Sorghum and Rice, while showing a positive association in Maize and Cowpea. Soil pH and soil K were non-significant predictors in Sorghum, and soil pH was also non-significant in Rice. Ranking crops by their overall sensitivity to environmental/soil variation (mean absolute standardised slope across all six predictors), Maize was the most responsive (648.9), followed by Sorghum (483.2), and Rice (401.7), with Cowpea (204.5) the least responsive.

Table 5. Standardised regression slopes by crop and predictor.

Crop	Rainfall	Temp.	Soil pH	Soil N	Soil P	Soil K
Maize	615.9***	−626.2***	−503.3***	1,129.9***	646.6***	371.3***
Sorghum	−1,018.4***	−74.0**	−24.4 (ns)	1,405.0***	138.8***	238.8***
Rice	−237.8***	−768.6***	−79.9 (ns)	582.8***	495.4***	245.5***
Cowpea	308.6***	−252.4***	185.1***	289.8***	188.8***	2.1 (ns)

**** $p < 0.001$, ** $p < 0.01$, ns = not significant.*

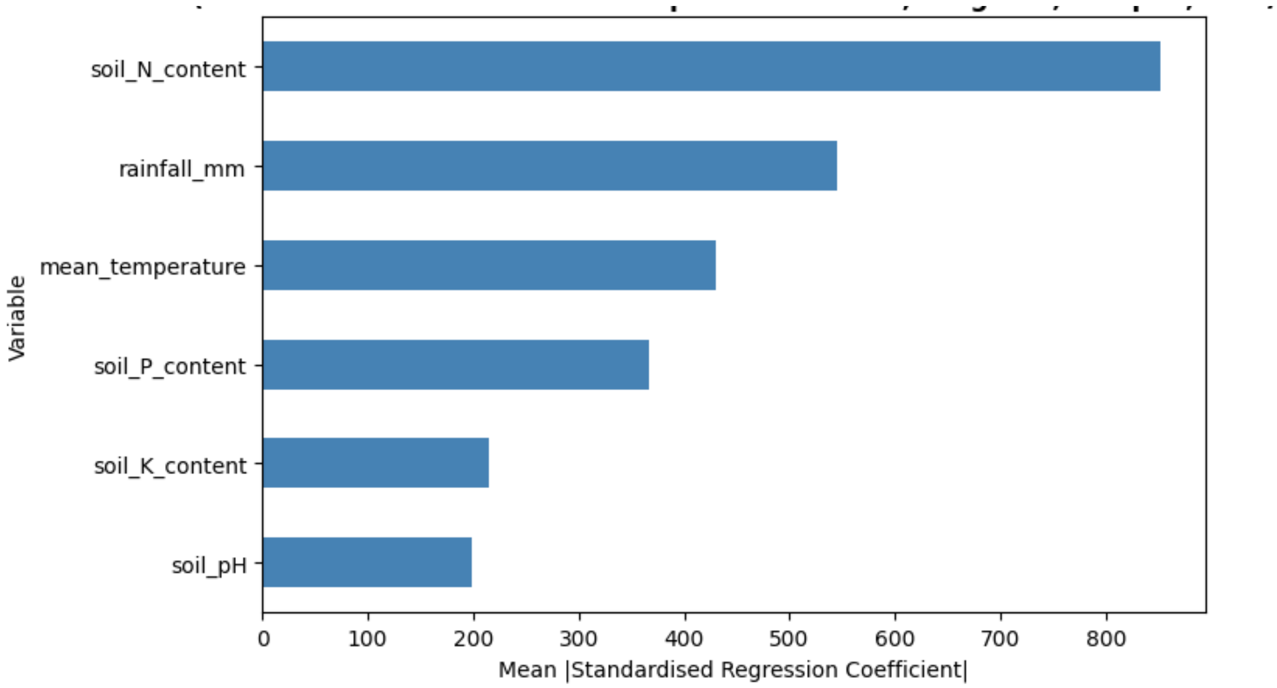

Figure 14: Relative Importance of Environmental and Soil Variables in Predicting Crop Yield

3.7 Correlations Between Environmental/Soil Variables and Grain Yield

Correlation analysis between grain yield and the six environmental/soil variables revealed distinct patterns across the four crops (Figure 14). Soil nitrogen content showed the strongest positive correlation with yield in Sorghum (r = 0.52) and Maize (r = 0.56), and a weaker but still positive correlation in Rice (r = 0.22) and Cowpea (r = 0.16). Mean temperature was negatively correlated with yield in all four crops, most strongly in Maize (r = −0.31) while in Sorghum the relationship was negligible (r = −0.03).

Rainfall showed an inconsistent relationship with yield across crops: positively correlated in Maize (r = 0.30), negligible in Rice (r = −0.09), and more strongly negative in Sorghum (r = −0.38). Soil pH showed weak correlations with yield across all crops, ranging from r = −0.25 (Maize) to r = 0.10 (Cowpea), none exceeding |0.25| except in Maize.

The six predictor variables were themselves highly intercorrelated in several crops. In Cowpea, rainfall and soil N content were almost perfectly correlated (r = 0.97), and soil P content was strongly correlated with both rainfall (r = 0.66) and soil N content (r = 0.74). In Sorghum, soil P and soil K content were very strongly correlated (r = 0.95), and mean temperature was strongly negatively correlated with soil K content (r = −0.88) and soil P content (r = −0.74). Similarly, in Rice, soil N and soil P content were strongly correlated (r = 0.95), and mean temperature was negatively correlated with soil N content (r = −0.86). In Maize, soil P and soil K content were moderately correlated (r = 0.74), and mean temperature was negatively correlated with soil P content (r = −0.71) and soil K content (r = −0.60).

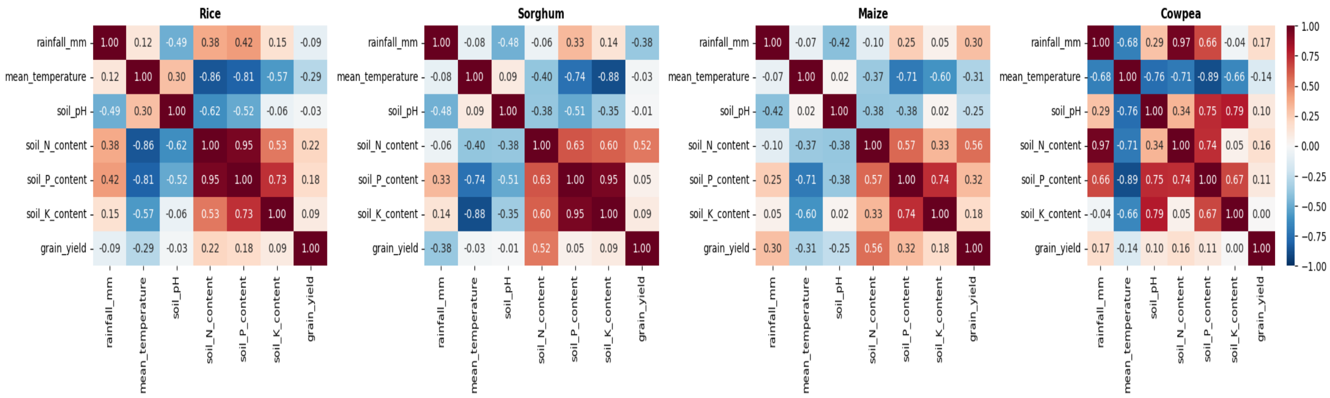

Figure 15: Correlation Heatmap: Yield vs Environmental/Soil Variables by Crop

4. Discussion

Hypothesis 1 that drought and Low-N stress significantly reduce grain yield in staple crops, with the magnitude of loss varying by country and genotype, was confirmed by the results from Maize and Sorghum. Both crops showed statistically significant and agronomically substantial yield reductions under both stresses, with Maize losing 72.5% of its yield under Drought and 87.0% under Low-N, and Sorghum losing 18.7% under Drought but 81.7% under Low-N. The Low-N loss observed for Maize is consistent with reports that Sub-Saharan Africa experiences among the most severe low-nitrogen-related yield penalties globally (80–90%) owing to intrinsically poor soil fertility compounded by limited fertiliser access among smallholder farmers (Amegbor et al. 2017, Aboderin et al., 2025). Sorghum's markedly different profile, combining strong drought tolerance with acute Low-N sensitivity is unsurprising given Sorghum's well-established reputation as a drought-resilient cereal, attributable to deeper rooting, more efficient stomatal regulation, and stay-green-mediated delayed senescence (Bankole et al., 2025; Hegde et al., 2023). That Low-N imposed the single largest yield penalty observed for either crop, exceeding even the severe Drought penalty in Maize, signals that nitrogen availability is the more fundamentally limiting constraint of the two in this dataset, overriding the species-level drought-tolerance advantage Sorghum otherwise displays. The substantial country-level variation in yield loss for both crops further confirms the location-dependent component of the hypothesis: some countries and, within them, some genotypes consistently suffered a disproportionately higher yield penalty than others, a distinction that matters practically for separating genuinely stress-tolerant material from material that merely performs well under favourable conditions

Hypothesis 2, that yield across staple crops responds to rainfall, temperature, and soil characteristics, with the strength of these relationships differing by crop, was confirmed by the results from the four crops. All six environmental and soil variables contributed to explaining yield variation, but their relative importance differed considerably among crops. Soil nitrogen content emerged as the most influential predictor in Maize, Sorghum, and Rice, and the second most important predictor in Cowpea, highlighting the central role of nitrogen availability in determining crop productivity across diverse agroecological conditions. The ranking of mean absolute standardised regression coefficients further demonstrated substantial crop-specific differences in environmental sensitivity, with Maize showing the greatest responsiveness to environmental and soil variation, followed by Sorghum and Rice, while Cowpea was considerably less responsive. This lower sensitivity of Cowpea may reflect its leguminous nature and ability to fix atmospheric nitrogen, making it less dependent on soil nutrient status than the cereal crops. Furthermore, the direction of the relationships also varied among crops; rainfall was positively associated with yield in Maize and Cowpea but negatively associated with yield in Sorghum and Rice, suggesting that excess rainfall may have constrained yields in some environments through factors such as waterlogging, reduced solar radiation, nutrient leaching, or increased disease pressure. The non-significant effects of soil pH and soil potassium in Sorghum, and of soil pH in Rice, also indicate that not all environmental variables exert the same influence across crops.

The findings of this study also have important implications for variety selection and deployment in the region. For Maize, the genotype clustering results identify SAMMAZ-52, ILOMAZ-77, ILOMAZ-61, ILOMAZ-83, and ILOMAZ-84 as the strongest candidates for deployment in stress-prone environments: the GGE biplot confirms these genotypes occupy a distinct, high-yielding region of genotype-by-environment space aligned with the Optimum-condition vectors across all five countries, while the cluster containing these genotypes also retained yield comparatively better than the lower-yielding majority cluster under both Drought and Low-N. This combination of high yield potential and above-average stress retention makes them the most defensible recommendation where breeding programmes must serve both favourable and stress-prone seasons within the same variety portfolio. For Sorghum, the highest-yielding cluster under Optimum conditions (Sekedo, S35, SAMSORG 52/53/54, Tegemeo, T27) is the same cluster the GGE biplot associates with the Drought environment vectors, indicating that these genotypes are clear deployment priority across the drought-prone locations. None of the Sorghum genotypes, however, showed a comparable advantage under Low-N, reinforcing that nitrogen-responsive breeding remains a distinct and unmet need even within this otherwise well-adapted genetic background. For Rice, IR 2793, FARO 40–44, IR 34686, and IR 36204 form the consistent high-yielding tier across both the clustering and biplot analyses, with WITA 1 standing out as a distinct, possibly niche-adapted genotype strongly associated with one specific country-year environment (Kenya, 2022) rather than broad adaptability. For Cowpea, TVu-1191, IT82D-716, and TVu-1218 emerge as the consistent top performers across both analyses, though the small size of this high-yielding tier relative to the much larger low-yielding majority suggests that the broader Cowpea germplasm evaluated here offers limited immediate options for yield improvement, and that expanding the pool of trialled genotypes should be a priority for future evaluation work in this crop.

5. Conclusions

The two hypotheses tested in this study were confirmed by the results. Drought and nitrogen stress were found to significantly reduce grain yield in staple crops across Sub-Saharan Africa, with the magnitude of loss varying by country and genotype. Nitrogen limitation emerged as the more universally binding constraint of the two stresses. All six environmental and soil variables examined in this study (rainfall, temperature, soil pH, soil N, soil P and soil K) contributed to explaining yield variation, but their relative importance differed considerably among crops, with Maize showing the highest responsiveness and Cowpea the least. At the genotype level, SAMMAZ-52, ILOMAZ-77, ILOMAZ-61, ILOMAZ-83, and ILOMAZ-84 were identified as the strongest Maize candidates for stress-prone deployment, while Sekedo, S35, and the SAMSORG line emerged as a rare case in Sorghum where high Optimum yield and drought resilience coincide in the same genotypes. IR 2793 and the FARO line were confirmed as the most broadly adapted high-yielding Rice genotypes, while TVu-1191, IT82D-716, and TVu-1218 represent the strongest available Cowpea material.

6. References

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